CS302P - Digital Logic Design (Practical) (Semester Quiz # 01)

Question # 4 of 10 (Start time: 04:06:15 PM, 11 November 2025)

If **Z = A' + B + C' + D** then what should be the values of A, B, C, and D so that the expression returns **Z = 0.**

Select the correct option

○	A = 1, B = 0, C = 0, D = 0
○	A = 1, B = 0, C = 1, D = 1
○	A = 0, B = 1, C = 0, D = 0
●	A = 1, B = 0, C = 1, D = 0

CS302P - Digital Logic Design (Practical) (Semester Quiz # 01)

Question # 3 of 10 (Start time: 04:06:00 PM, 11 November 2025)

In Electronics Workbench, to use a **Connector** component in circuit, we open _____ toolbox.

Select the correct option

○	Basic
○	Logic Gates
○	Instruments
●	Miscellaneous

CS302P - Digital Logic Design (Practical) (Semester Quiz # 01)

Question # 1 of 10 (Start time: 04:05:06 PM, 11 November 2025)

According to De-Morgan's Law, the expression **(A+BC)'** is equivalent to _____.

Select the correct option

●	A'B'+A'C'
○	A' + B' + C'
○	A'B+AC'
○	A'B'+BC

CS302P - Digital Logic Design (Practical) (Semester Quiz # 01)

Question # 2 of 10 (Start time: 04:05:48 PM, 11 November 2025)

We can change the number of inputs for a gate by using _____.

Select the correct option

◉	Gate Properties
○	Gate Diagram
○	Gate Structure
○	Truth Table

CS302P - Digital Logic Design (Practical) (Semester Quiz # 01)

Question # 10 of 10 (Start time: 03:49:38 PM, 11 November 2025)

According to the distributive law of Boolean algebra, which of the following statements is **TRUE**?

Select the correct option

◉	X + (Y Z) = (X + Y) (X + Z)
○	X + (Y Z) = X+Y+Z
○	X + (Y Z) = XYZ
○	X + (Y Z) = XY+XZ

CS302P - Digital Logic Design (Practical) (Semester Quiz # 01)

Question # 9 of 10 (Start time: 03:49:26 PM, 11 November 2025)

A commutative law of boolean algebra involves _____ boolean variable(s).

Select the correct option

○	1
○	3
○	4
◉	2

CS302P - Digital Logic Design (Practical) (Semester Quiz # 01)

Question # 7 of 10 (Start time: 03:49:01 PM, 11 November 2025)

For inverting an input, we use _____ gate.

Select the correct option

○	AND Gate
◉	NOT Gate
○	OR Gate
○	XOR Gate

CS302P - Digital Logic Design (Practical) (Semester Quiz # 01)

Question # 6 of 10 (Start time: 03:48:48 PM, 11 November 2025)

_____ gate gives high output (1) if all the inputs are 1's. otherwise the output will be low (0)

Select the correct option

○	OR
○	NOT
○	NOR
◉	AND

CS302P - Digital Logic Design (Practical) (Semester Quiz # 01)

Question # 8 of 10 (Start time: 03:49:13 PM, 11 November 2025)

To draw a logic circuit for boolean expression **AB'+C**, how many gates are used?

Select the correct option

○	2
○	1
○	4
●	3

CS302P - Digital Logic Design (Practical) (Semester Quiz # 01)

Question # 4 of 10 (Start time: 03:48:17 PM, 11 November 2025)

The Boolean function **Z = A + B . C'** can be implemented in digital logic circuit by connecting _____ gates together.

Select the correct option

○	4
○	1
○	3
●	2

CS302P - Digital Logic Design (Practical) (Semester Quiz # 01)

Question # 5 of 10 (Start time: 03:48:28 PM, 11 November 2025)

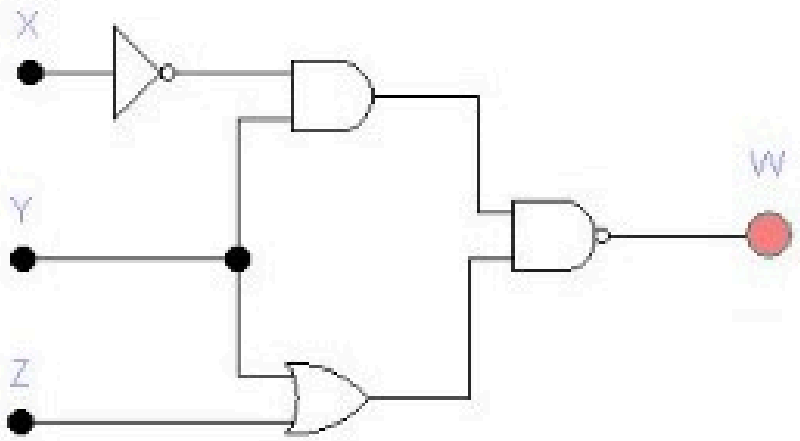
DE Morgan's Theory is used to convert _____ gates to _____ones.

Select the correct option

○	OR/NOT, NOT/OR
◉	AND/OR, NOR/NAND
○	AND/XOR, XOR/AND
○	AND/NAND, OR/NOR

Question # 2 of 10 (Start time: 03:47:49 PM, 11 November 2025)

What will be the Simpilifed Boolean Expression of the given Gate Level Diagram.



Select the correct option

○	A + C'
○	A + B'
●	A + B' + C
○	B + C

CS302P - Digital Logic Design (Practical) (Semester Quiz # 01)

Question # 10 of 10 (Start time: 04:07:47 PM, 11 November 2025)

For inverting an input, we use _____ gate.

Select the correct option

○	XOR Gate
○	AND Gate
○	OR Gate
●	NOT Gate

CS302P - Digital Logic Design (Practical) (Semester Quiz # 01)

Question # 1 of 10 (Start time: 03:47:32 PM, 11 November 2025)

In EWB, logic 1 is represented by using the _____ line

Select the correct option

○	V_{CC}
●	V_{DD}
○	-V
○	Ground

CS302P - Digital Logic Design (Practical) (Semester Quiz # 01)

Question # 3 of 10 (Start time: 03:48:04 PM, 11 November 2025)

Which version of Electronics Workbench is being used in this course for creating and simulating digital circuits?

Select the correct option

○	version 12.5
●	version 6
○	version 5.12
○	version 10

CS302P - Digital Logic Design (Practical) (Semester Quiz # 01)

Question # 9 of 10 (Start time: 04:07:34 PM, 11 November 2025)

Click the _____ to turn power on. Click switch again to turn power off.

Select the correct option

◉	Power Switch
○	Simulation Switch
○	Control Switch
○	Circuit Switch

CS302P - Digital Logic Design (Practical) (Semester Quiz # 01)

Question # 8 of 10 (Start time: 04:07:17 PM, 11 November 2025)

If A = 0 & B = 1, What will be the output of F = (A. B) + A' (A+B).

Select the correct option

●	1
○	0

CS302P - Digital Logic Design (Practical) (Semester Quiz # 01)

Question # 7 of 10 (Start time: 04:07:05 PM, 11 November 2025)

A commutative law of boolean algebra involves _____ boolean variable(s).

Select the correct option

○	3
○	1
○	4
◉	2

CS302P - Digital Logic Design (Practical) (Semester Quiz # 01)

Question # 6 of 10 (Start time: 04:06:50 PM, 11 November 2025)

DE Morgan's Theory is used to convert _____ gates to _____ones.

Select the correct option

○	AND/XOR, XOR/AND
◉	AND/OR, NOR/NAND
○	OR/NOT, NOT/OR
○	AND/NAND, OR/NOR

CS302P - Digital Logic Design (Practical) (Semester Quiz # 01)

Question # 5 of 10 (Start time: 04:06:39 PM, 11 November 2025)

In Electronics Workbench, the default value for the duty cycle of the clock source is _____

Select the correct option

●	50%
○	2%
○	100%
○	5%